LEVEL SETS OF THE INTEGRAL OF A WEIGHTING ARE LOCALLY CONSTRUCTIBLE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 22625-22669, immediately following the statement at 22618-22623.

Editorial note. The source line 22629 contains the author's typo 'weighthing'; it is retained verbatim inside the counted proof.

Editorial note. Four cross-chapter pointers (Coherent, Morphisms, Properties) plus four same-chapter weightings lemmas are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Level sets of a weighting are locally constructible. chapter lemma lemma-jumps-w, cited at proof line 22662. Transfers constructibility from w to the sum function in the trivial-residue-extension case.
- Base change of the integral and of weightings. chapter lemmas lemma-weighting-check-after-etale-base-change and lemma-weighting-base-change, cited at proof lines 22626-22629. Shows the function of interest commutes with arbitrary base change, which licenses the surjective finite presentation reduction.
- Chevalley's theorem and massaging a finite morphism. Morphisms, theorem morphisms-theorem-chevalley and lemma morphisms-lemma-massage-finite, cited at proof lines 22656-22667. Makes images of constructible sets constructible and reduces to trivial residue field extensions.
- Stratification by the finite value set of w. Properties, lemma properties-lemma-stratification-locally-finite-constructible, cited at proof lines 22664-22665. Gives that w takes only finitely many values, so the integral is a finite sum of constructible pieces.
- Induction principle for quasi-compact quasi-separated schemes. Coherent, lemma coherent-lemma-induction-principle, cited in the proof. Supports the reduction of the ambient scheme to a manageable piece.

Source reference. Original reference 08DR: focused-seeds/geometry/S9-morph-jumps-int-w/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F3G: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22514–22514. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05K0: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12457–12457. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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Source reference. Original lemma 0F3B: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22173–22173. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F39: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22111–22111. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03HW: focused-seeds/geometry/S9-morph-jumps-int-w/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 054K: focused-seeds/geometry/S9-morph-jumps-int-w/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0F2M: focused-seeds/geometry/S9-morph-jumps-int-w/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a locally quasi-finite morphism of finite presentation. Let $w : X \to \mathbf{Z}$ be a weighting of f. Then the level sets of the function $\int_f w$ are locally constructible in Y.*

Proof. By Lemma 0F39 formation of the function $\int_f w$ commutes with arbitrary base change and by Lemma 0F3B after base change we still have a weighthing. This means that if we can find $Y' \to Y$ surjective and of finite presentation, then it suffices to prove the result after base change to Y', see Morphisms, Theorem 054K.

The question is local on Y hence we may assume Y is affine. Then X is quasi-compact and quasi-separated (as f is of finite presentation). Suppose that $X = U \cup V$ are quasi-compact open. Then we have

$$\int_f w = \int_{f|_U} w|_U + \int_{f|_V} w|_V - \int_{f|_{U\cap V}} w|_{U\cap V}$$

Thus if we know the result for $w|_U$, $w|_V$, $w|_{U\cap V}$ then we know the result for w. By the induction principle (Cohomology of Schemes, Lemma 08DR) it suffices to prove the lemma when X is affine.

Assume X and Y are affine. We may choose an open immersion $X \to T$ where $T \to Y$ is finite, see Lemma 05K0. Because we may still base change with a suitable $Y' \to Y$ we can use Morphisms, Lemma 03HW to reduce to the case where all residue field extensions induced by the morphism $T \to Y$ (and a foriori induced by $X \to Y$) are trivial. In this situation $\int_f w$ is just taking the sums of the values of w in fibres. The level sets of w are locally constructible in X (Lemma 0F3G). The function w only takes a finite number of values by Properties, Lemma 0F2M.

Hence we conclude by Morphisms, Theorem 054K and some elementary arguments on sums of integers. □